



Facebook's Internet-providing aircraft Aquila waiting to take-off on its test flight (Source: Facebook.com)

Realising the potential of a connected population, Internet majors such as Facebook, Microsoft and Google are all eager to pitch in with their innovative technologies to connect rural India. But, the Indian government is being selective and careful in accepting proposals. For example, Facebook's initial attempt to connect rural India through the Free Basics program was stopped by the Telecom Regulatory Authority of India (TRAI) as it allowed access only to certain content, in contrast with the principles of net neutrality. Recently, Microsoft's attempt to use whitespace (unused television bandwidth) for connectivity in Maharashtra also ran into rough weather because national security agencies thought it was risky to use an unregulated band for last-mile connectivity.

However, enthusiasm has not ebbed and we see these and other companies coming up with fresh ideas! Let us have a look at the many options that are being considered in India and globally to connect remote areas affordably.

Satellites to the rescue

Satellite communication is one of the most effective methods of connecting remote areas. Why? Patricia Cooper, vice president of satellite government affairs at SpaceX, probably explained it best while addressing a US Senate Committee last year. She told that

use of a space-based network meant companies didn't need to install, rip up and reinstall cabling in order to provide a service.

"In other words, the common challenges associated with siting, digging trenches, laying fibre and dealing with property rights are materially alleviated through a space-based broadband network," Cooper said. This improves reach and reduces cost in the long run, even though the initial setup cost might seem high.

Companies like O3b Networks, OneWeb, SpaceX and Boeing are working towards placing satellite constellations in the low-Earth and medium-Earth orbits so as to reach hitherto unconnected regions in a cost-effective and efficient way.

Recently, Boeing Satellite Systems International entered into a deal to build seven communication satellites for Luxembourg-based SES' O3b Networks—a project that aims to connect the other three billion people in the world to the Internet. Currently, the O3b constellation has 12 first-generation satellites in the medium-Earth orbit, around 8000km above the Earth. The new satellites being built by Boeing are expected to launch in 2021, and bring 1-terabit Internet connectivity through the O3b network. The entire constellation is expected to cover four-fifths of the Earth's surface.

Boeing is also working towards a 1000+ satellite constellation that will

MECO®

SINCE 1962

TRMS DIGITAL MULTIMETERS



171B+TRMS

153B+TRMS

3% Digit 6000 Count LCD with Backlight

- Auto Ranging
- 600mV ~ 1000V DC
- 600mV ~ 750V AC TRMS
- 600µA ~ 20A DC
- 600µA ~ 20A AC TRMS
- 600Ω ~ 60MΩ
- 9.999nF ~ 9.999mF
- 99.99Hz ~ 20.00MHz
- 0.1% ~ 99.9% Duty Cycle
- -40°C ~ 1000°C (-40°F ~ 1832°F) (171B+TRMS)
- K Type Thermocouple (Upto 260°C) (171B+TRMS)
- REL Δ, Max/Min (171B+TRMS)
- Diode Test, Data Hold, Audible Continuity, Auto Power off & Low Battery Indicator

Meco Meters Private Limited

Plot No. EL-80, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400719 (INDIA)

Tel : 091-22-2767 3300 (Board), 2767 3311-16 (Sales)

Fax : 091-22-27673310, 27673330

Email : sales@mecoinst.com

Web : www.mecoinst.com